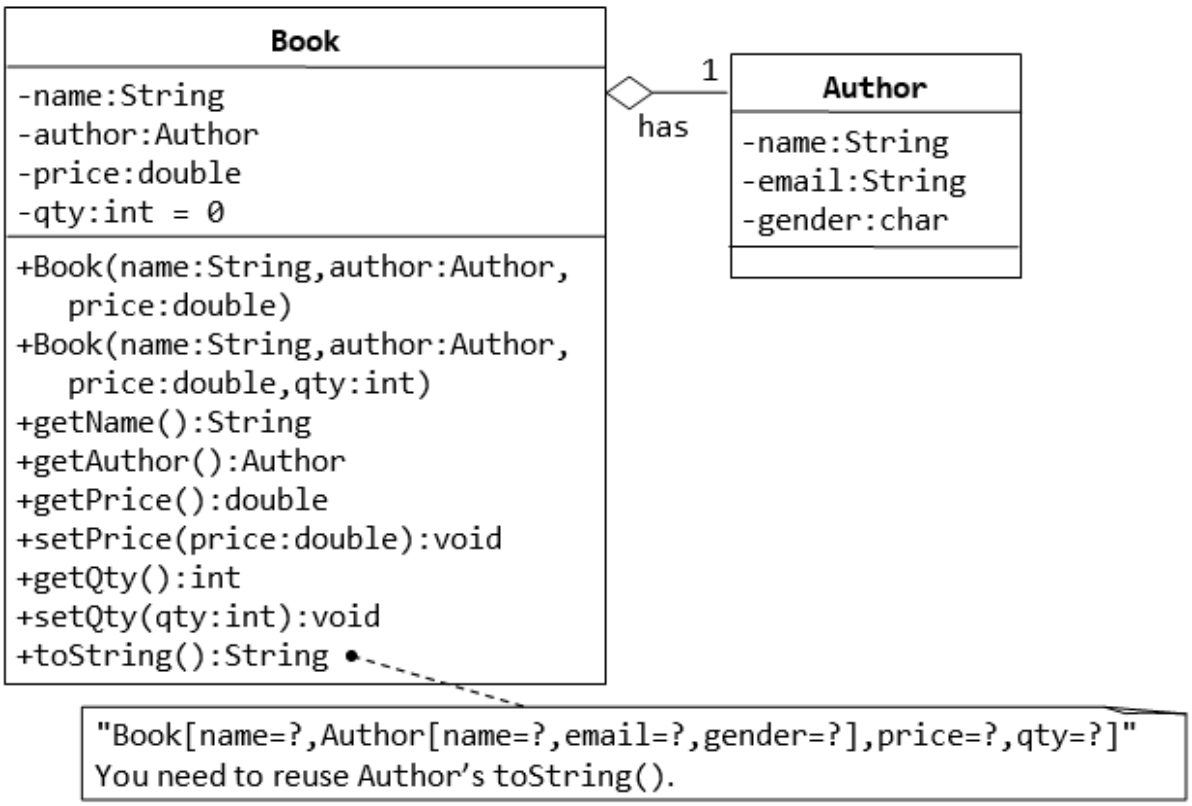
1. Design and implement Employee Information System (EIS), comprising of a class called Employee which includes details such as Employee ID, Name, Designation, Salary. The class also provides functionalities such as displaying employee details, searching for an employee based on employee ID, provision for salary increment (10%), etc. The main class of EIS initially collects all employee details and later it provides menu items such as search, salary increment, and display.

2. Develop Stack with following operations
   i) createNew( ) ii) push( ) iii) pop( ) iv) display( ) v) empty( )

3.



A class called Book is designed (as shown in the class diagram) to model a book written by *one* author. It contains:

- Four private instance variables: name (String), author (of the class Author you have just created, assume that a book has one and only one author), price (double), and qty (int);
- Two constructors:

```
public Book (String name, Author author, double price) { ..... }
public Book (String name, Author author, double price, int qty) { ..... }
```

- public methods getName(), getAuthor(), getPrice(), setPrice(), getQty(), setQty().
- A toString() that returns "Book[name=?, Author[name=?,email=?, gender=?], price=?, qty=?". You should reuse Author's toString().

Write the Book class (which uses the Author class). Also write a test driver called TestBook to test all the public methods in the class Book. Take Note that you have to construct an instance of Author before you can construct an instance of Book.

4. A book shop maintains the inventory of books that are being sold at the shop. The list includes details such as author, title, price, publisher and stock position. Whenever a customer wants a book, the salesperson inputs the title and author, and the system searches the list and displays whether it is available or not. If it is not, an appropriate message is displayed. If it is, then the system displays the book details and requests for the number of copies required. If the requested copies are available, the total cost of the requested copies is displayed; otherwise, the message "required copies not in stock" is displayed. Design a system using a class called books with suitable member functions and constructors

5. Develop a java application for an automobile shop which sells used cars. The application maintains details about the car such as name, model, year of purchase, color, and price.

   Maintain a database for each of the car's objects. For example: find the initial 5 car objects details
   1. "Brezza", "VX", "02/10/2017", "White", INR 6,50,500.00
   2. "Venue", "SX", "20/9/2018", "White", INR 7,50,000.00
   3. "Ciaz", "VX", "15/12/2015", "White", INR 6,00,000.00
   4. "Ecosport", "LX", "21/6/2018", "Blue", INR 5,00,500.00
   5. "Nexon", "SX", "15/10/2019", "Green", INR 8,50,500.00

   Application automatically removes the car details from the database once the booking is done.
   Define a class "Car", include all data members and methods. Include getter, setter functions for all the data members. In order to provide the car details, override the function toString(). Write a testing class (main class) to validate each function.